

Chapter 10

Liquids and Solids

Chapter Outline

[10.1 Intermolecular Forces](#)

[10.2 Properties of Liquids](#)

[10.3 Phase Transitions](#)

[10.4 Phase Diagrams](#)

[10.5 The Solid State of Matter](#)

[10.6 Lattice Structures in Crystalline Solids](#)

CH 10.2 I made a concept video, but it's optional (you will not be tested on this content)

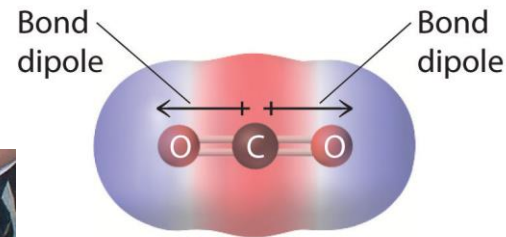
Sections 10.5/10.6 Will not be covered in CHEM 120.



Marie-Claude Paquette throwing the boiling water into the -30 centigrade air creating beautiful designs (St. Agathe, QC)

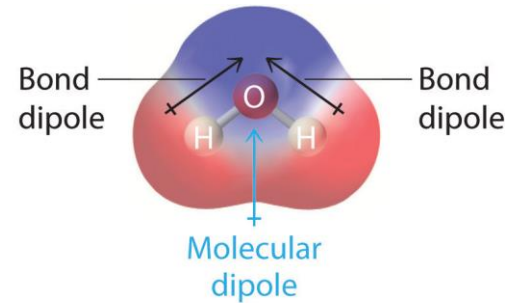
Phase Diagrams

- Different substances have very different conditions where phases will be stable.
- Phase diagrams consolidate this information



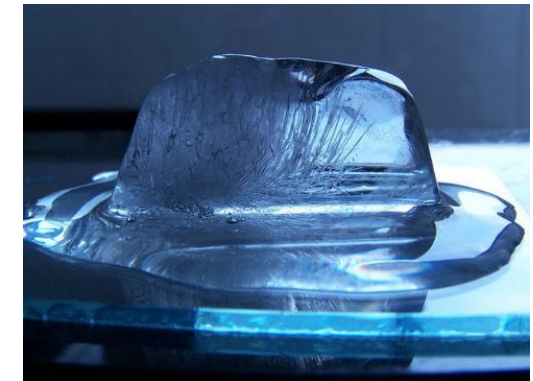
(a) No net dipole moment

- **dry ice**
- **frozen CO_2**
- **-79°C**
- **sublimates**

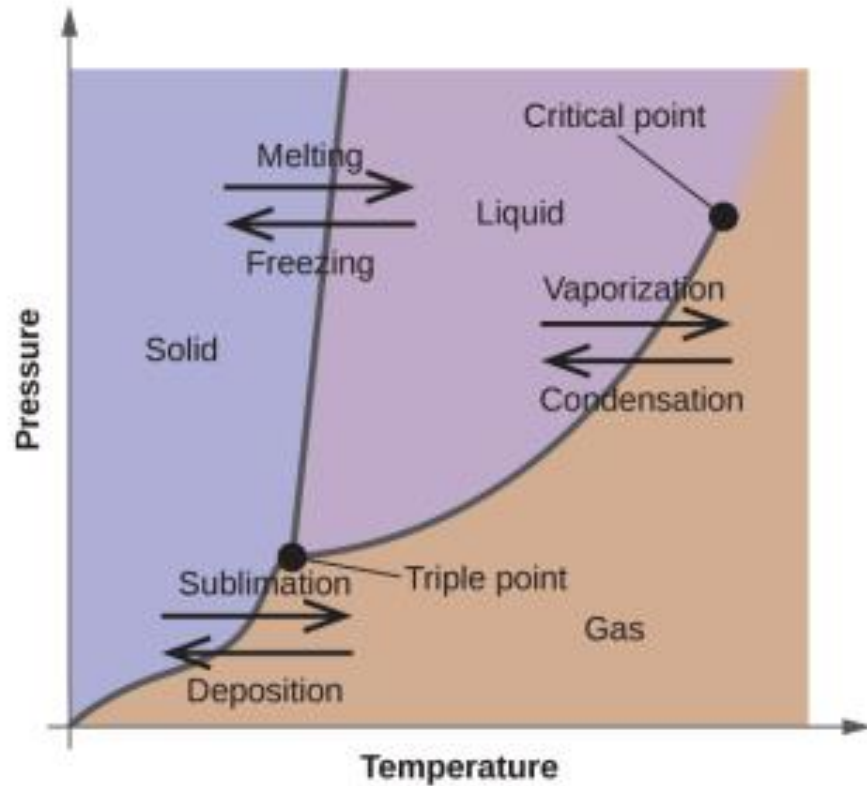


(b) Net dipole moment

- **ice**
- **frozen H_2O**
- **0°C**
- **melts**

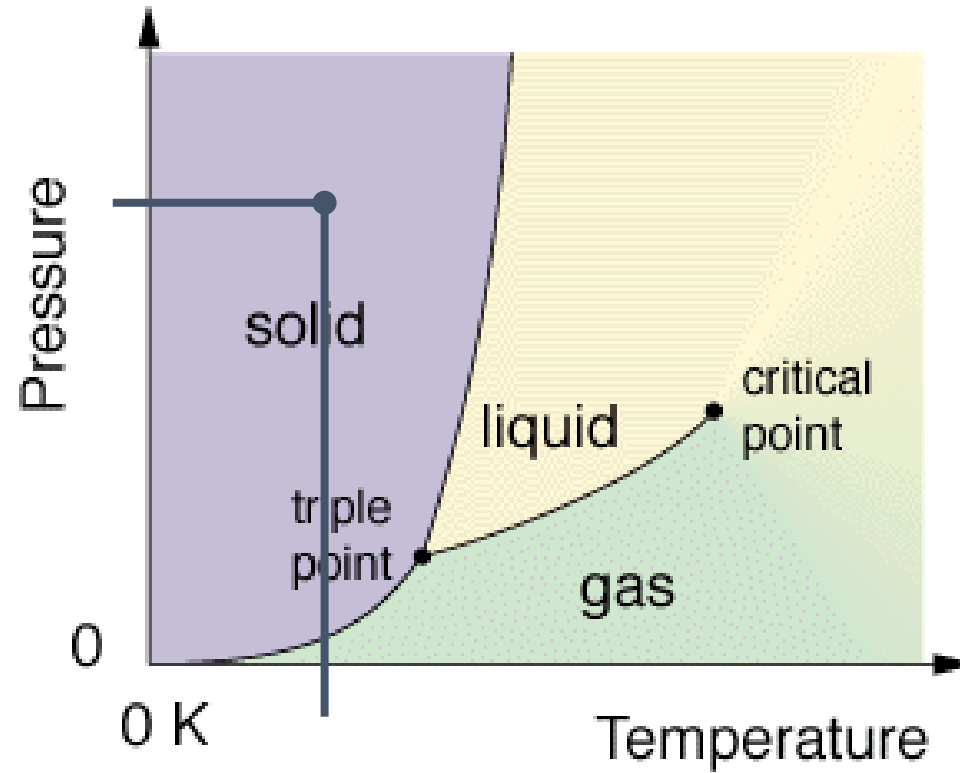


Phase Diagrams



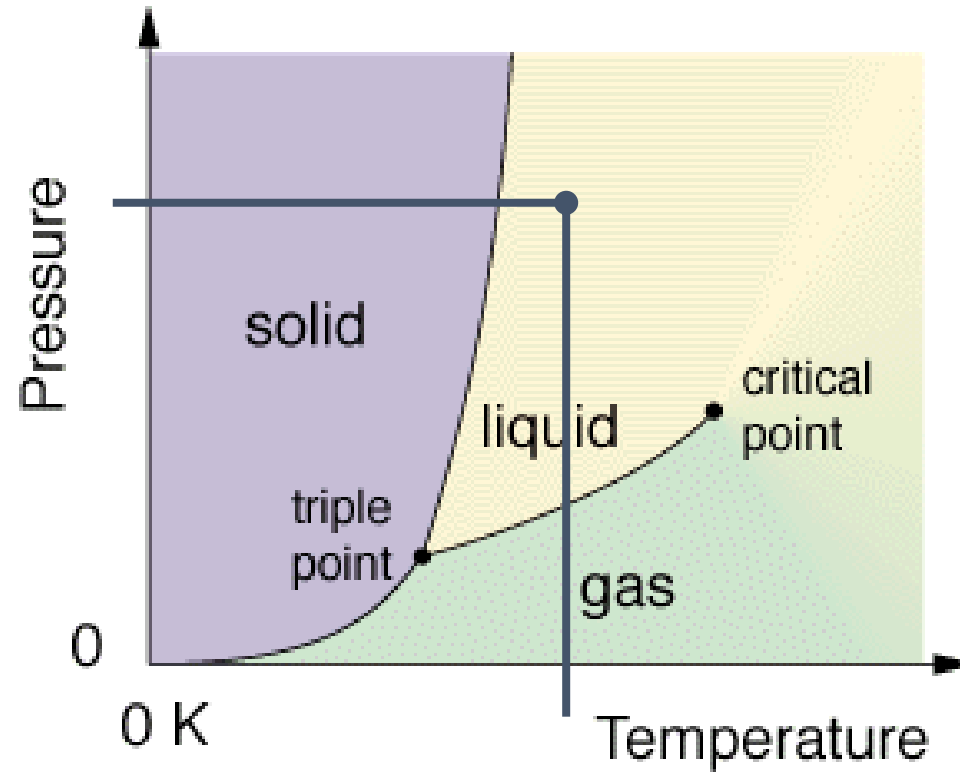
- Solid, liquid, or gas for a given temperature and pressure?
- Different for every substance.

Phase Diagrams



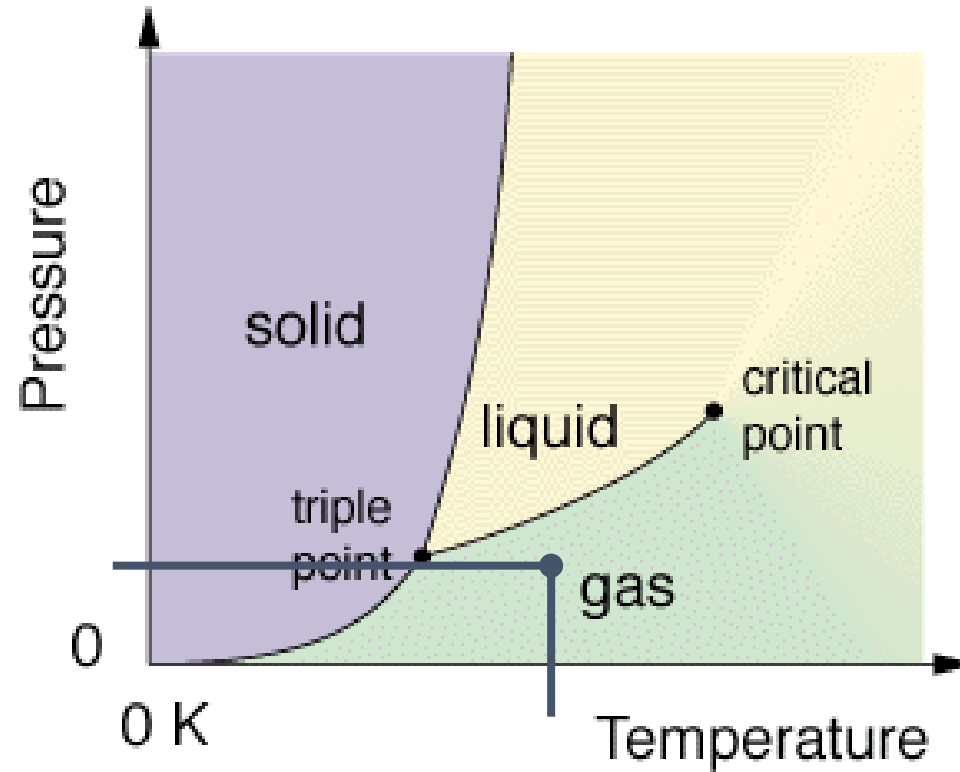
- At this temperature and pressure, the substance is solid, at equilibrium.

Phase Diagrams



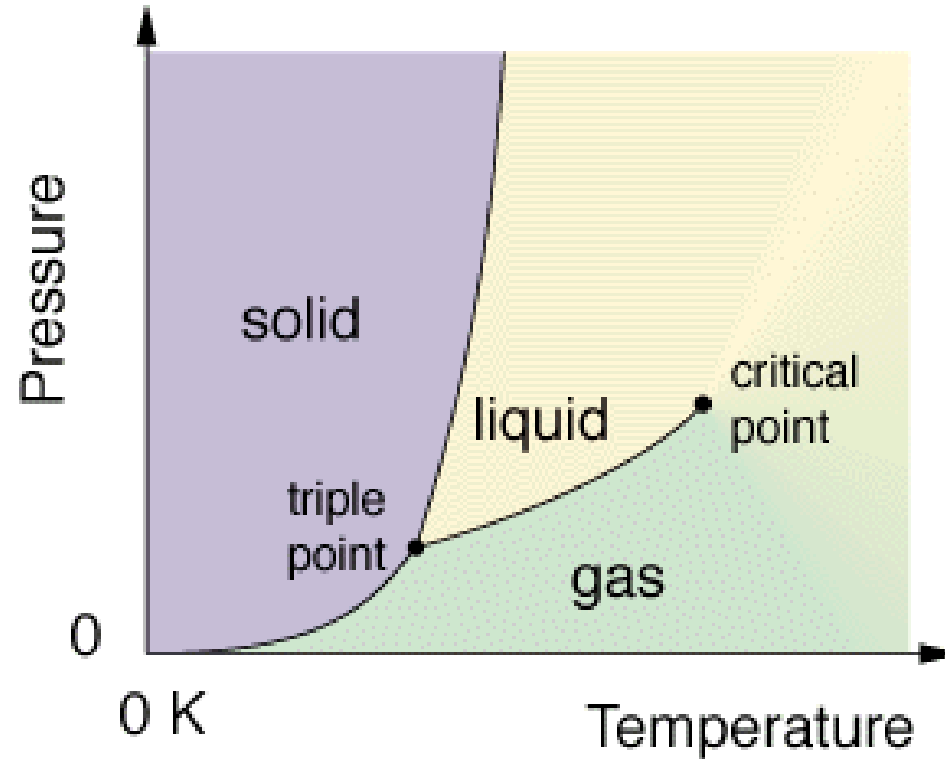
- At this temperature and pressure, the substance is liquid, at equilibrium.

Phase Diagrams



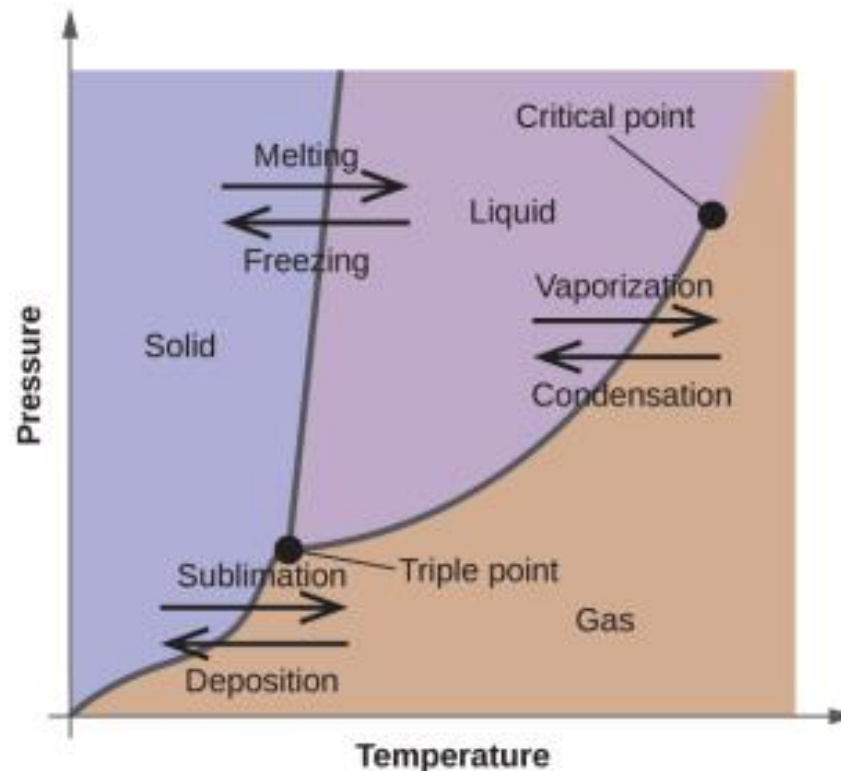
- At this temperature and pressure, the substance is gaseous, at equilibrium.

Phase Transitions



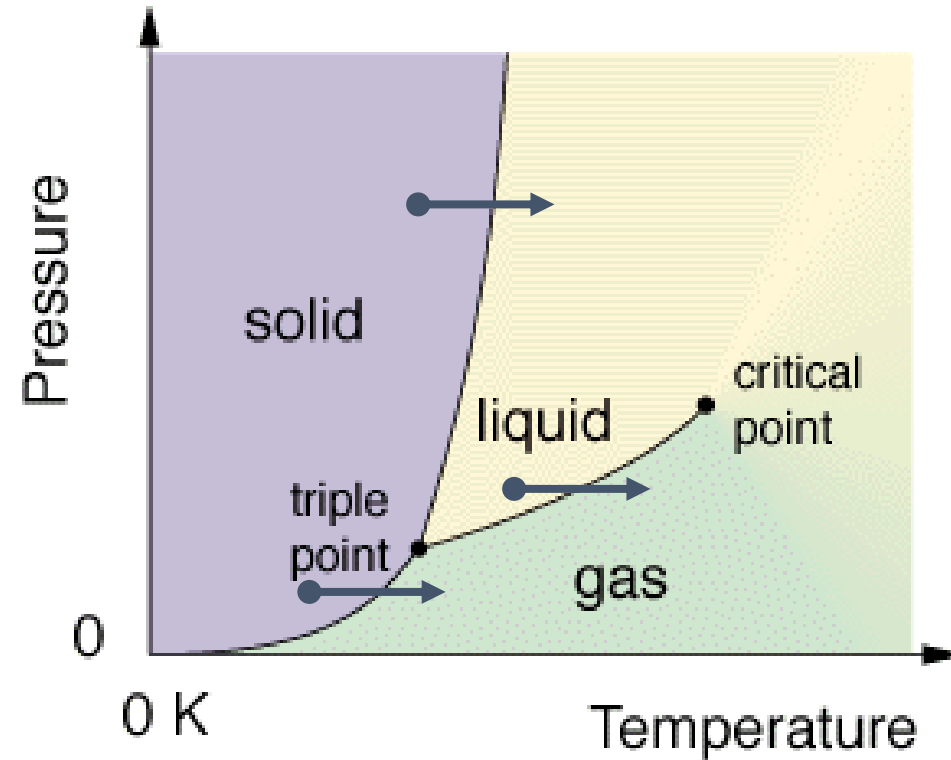
- We can learn a lot about phase transitions just by inspecting the phase diagram.
- The lines show combinations of P, T where the transition between two phases occur (coexistence curve) at equilibrium conditions.

Lines between Regions



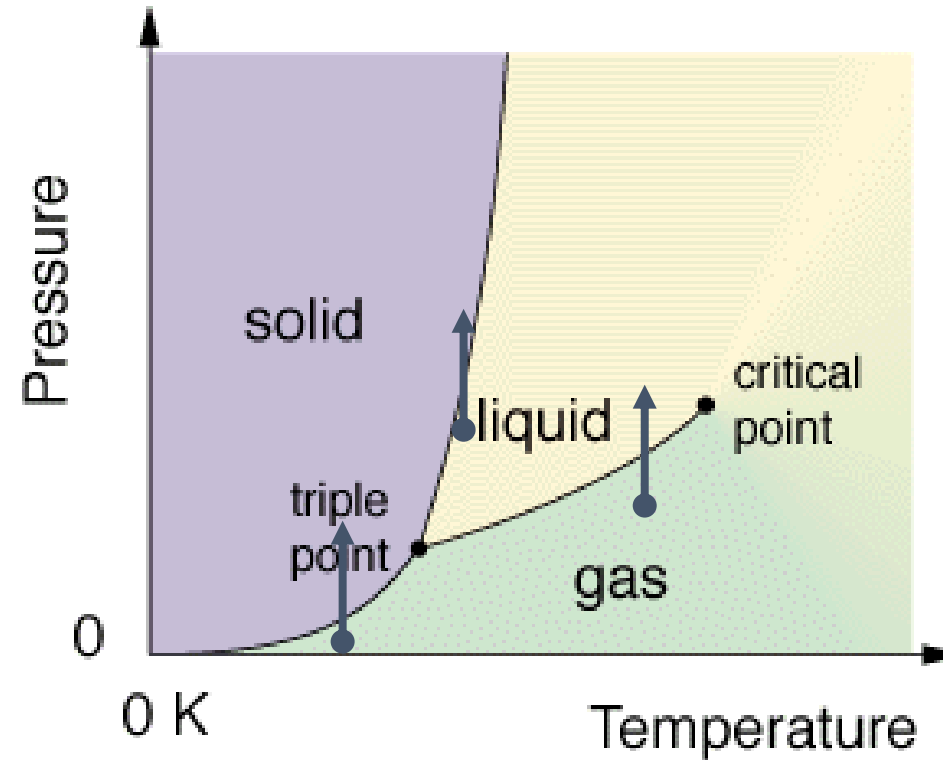
If we change P & T , we can cause the system to cross phase boundaries.

Phase Transitions



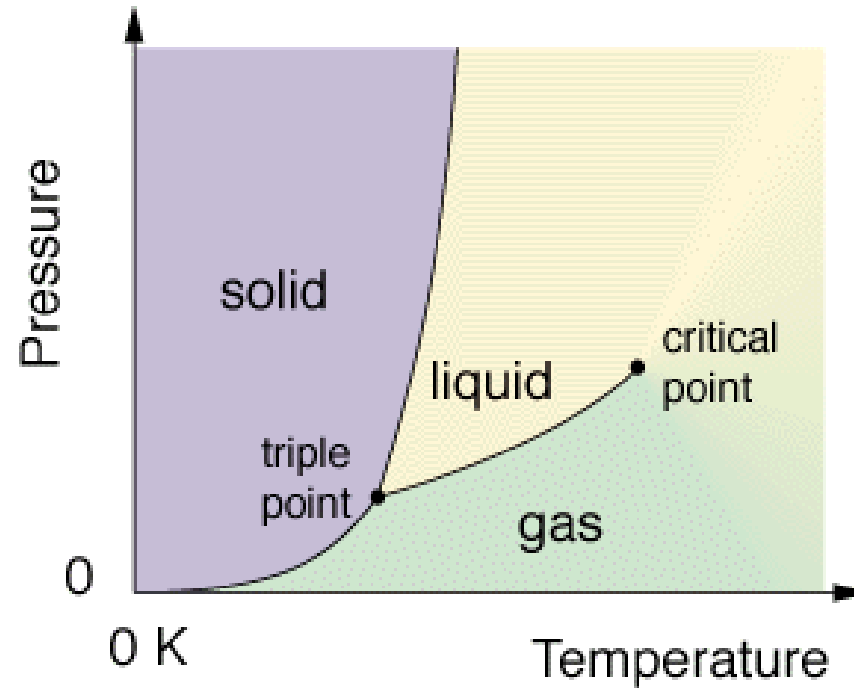
- Moving over a transition at constant P , increasing T :
 - horizontal, left to right
- Enthalpy increases (endothermic-put heat in to raise T), Entropy increases also.

Phase Transitions



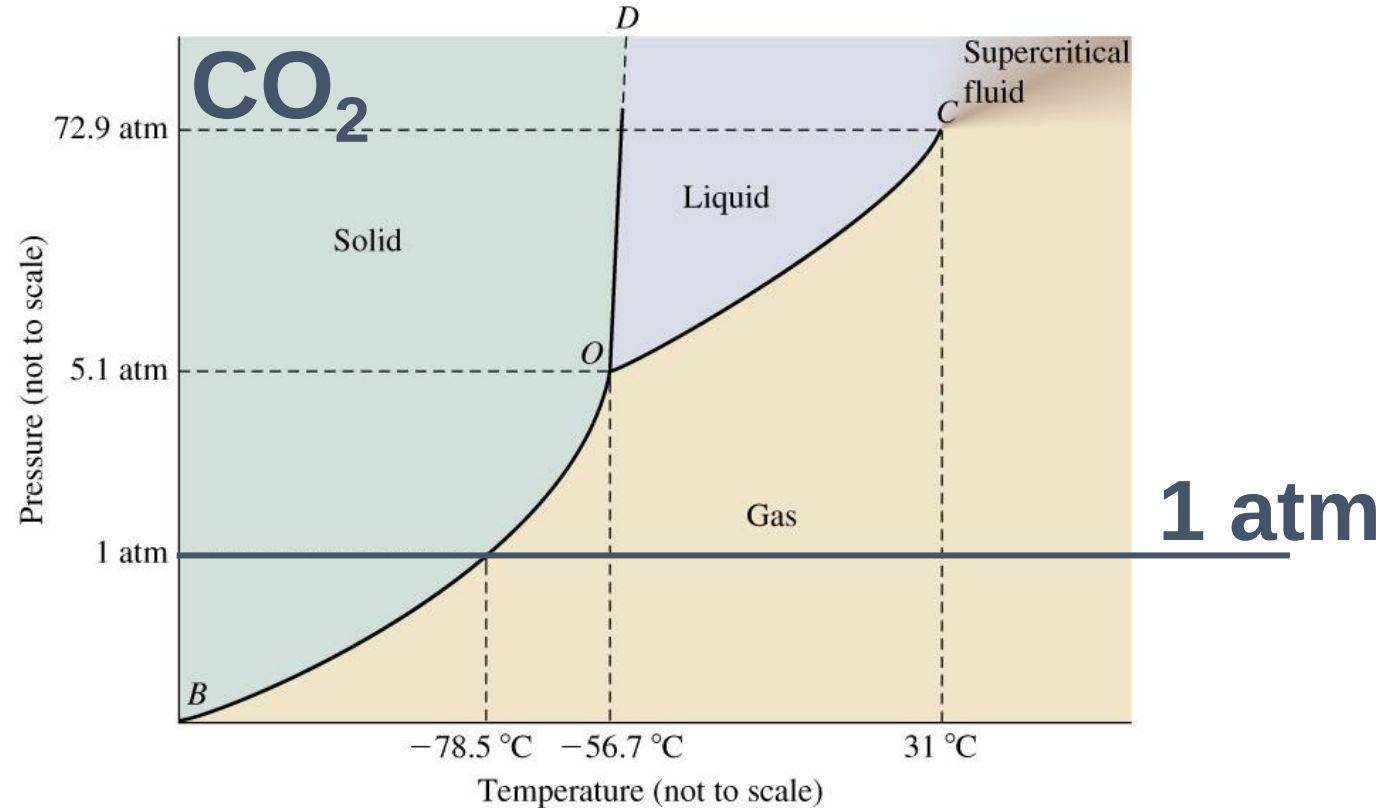
- Moving over a transition at constant T , increasing P :
 - vertical, bottom to top
- Density increases

Phase Diagrams



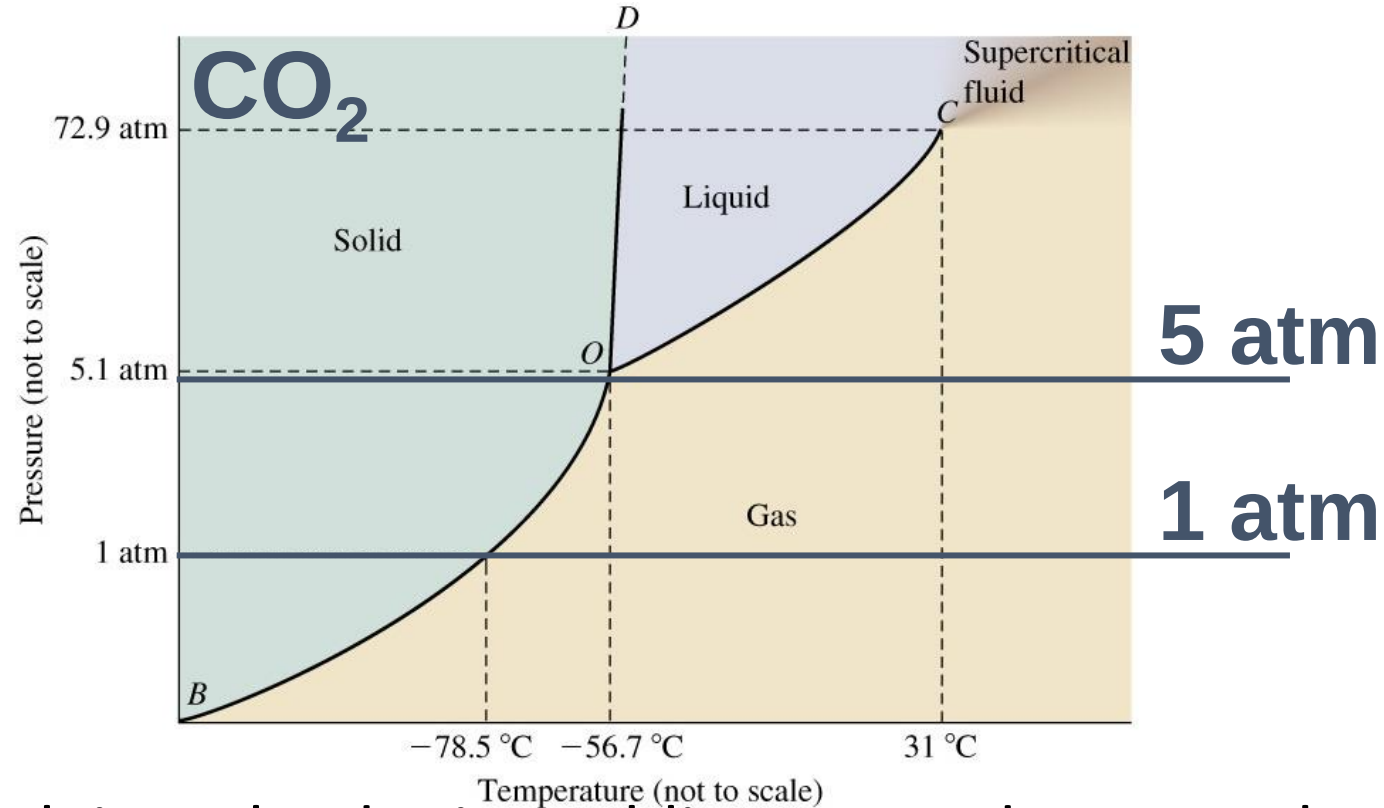
- The three phases can only co-exist in equilibrium at a single temperature and pressure: the triple point. Used for thermometer calibration.
- Pressures **BELOW** the triple point pressure, the liquid phase is never the stable form at any temperature.

Phase Diagrams



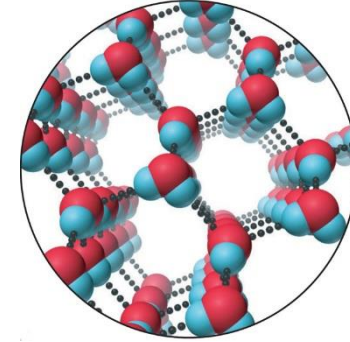
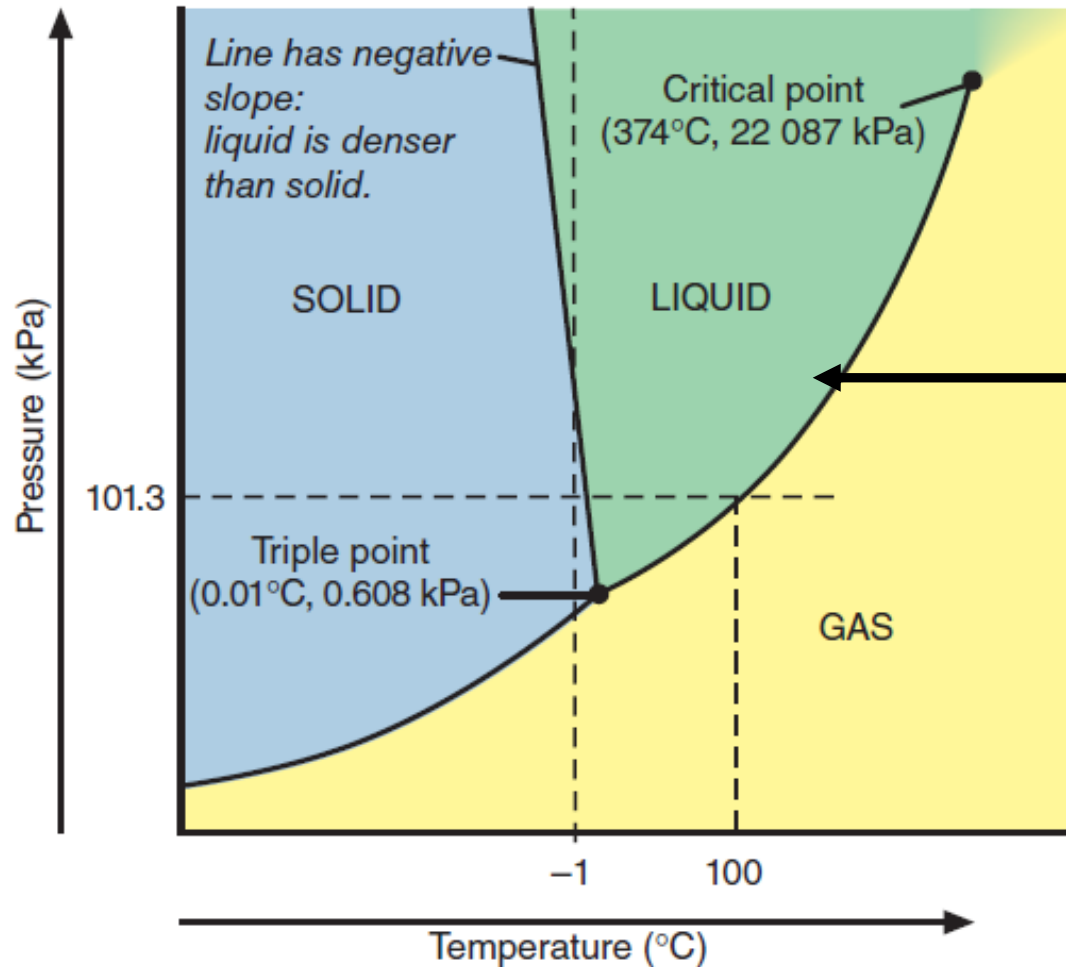
- As you heat CO_2 from below -79°C to room temperature (1 atm), you cross the sublimation curve: solid to gas transition.

Phase Diagrams

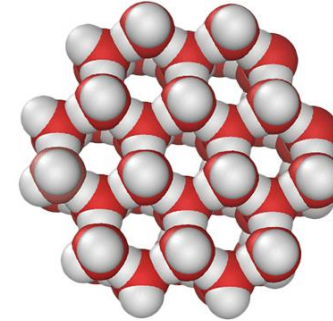
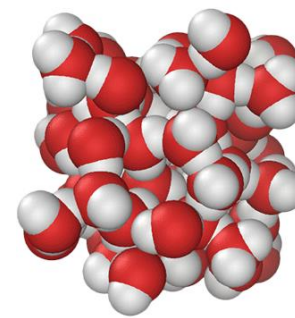


- This explains why dry ice sublimates under normal conditions rather than melting.
- To get liquid CO₂ you need pressures greater than 5 atmospheres.
- For CO₂, you can form the solid from the liquid by applying pressure → Solid is denser than liquid.

Phase Diagrams

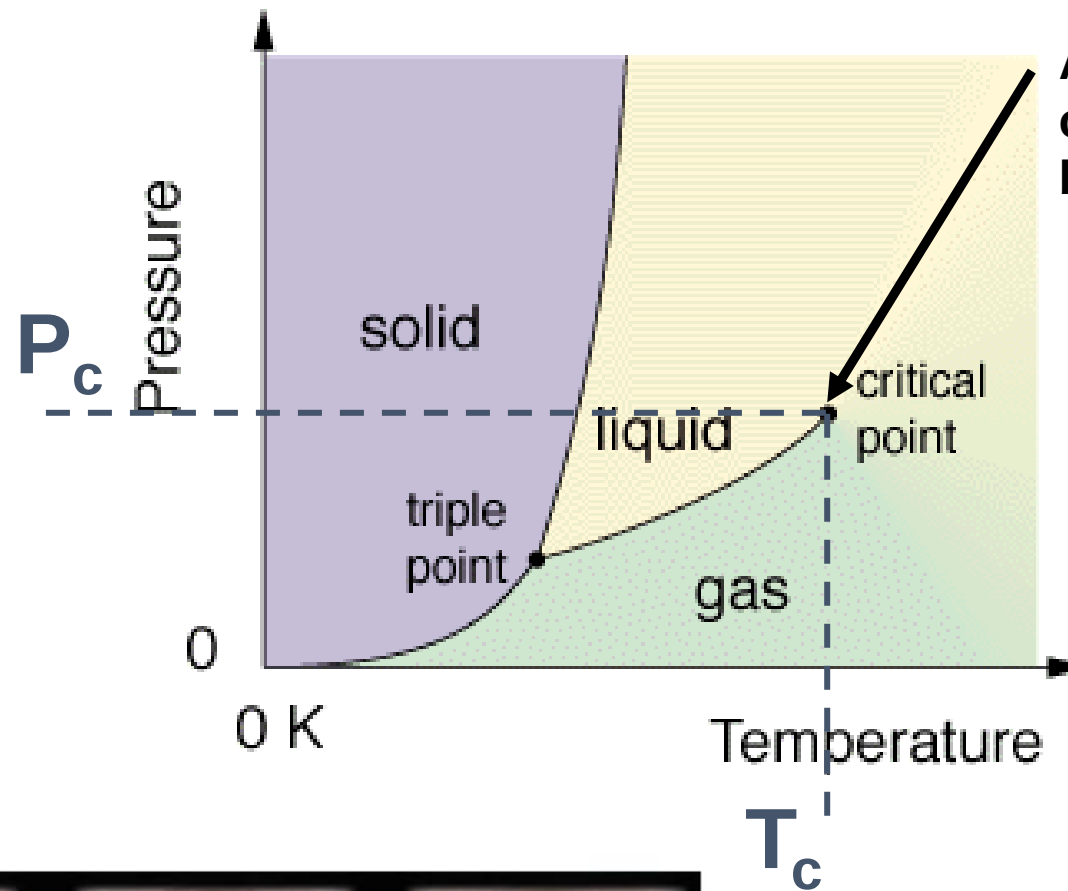


The solid-liquid line slants to the *left* for H₂O, because the solid is *less* dense than the liquid. It is known that water expands on freezing.



- heating at 1 atm pressure, gives solid to liquid to gas.
- For water, you can form liquid from solid by applying pressure- liquid is denser.

Critical Point



At the critical point, the densities of the liquid and gas phases become equal.

- The line separating liquid and gas phases ends at the critical point.

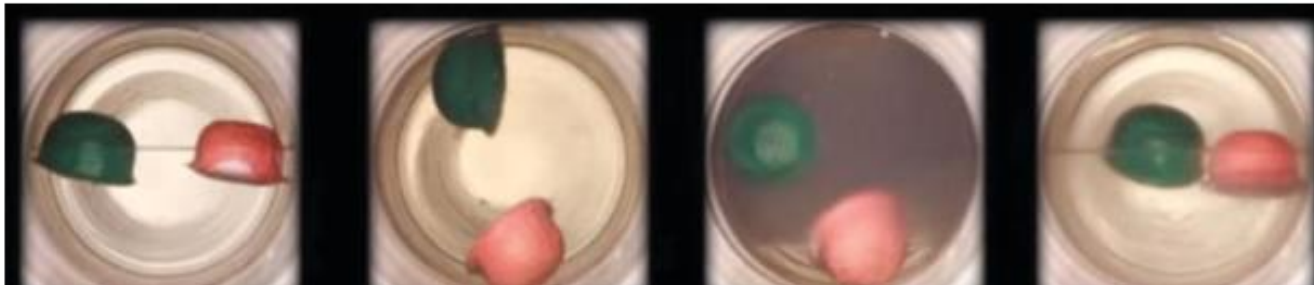
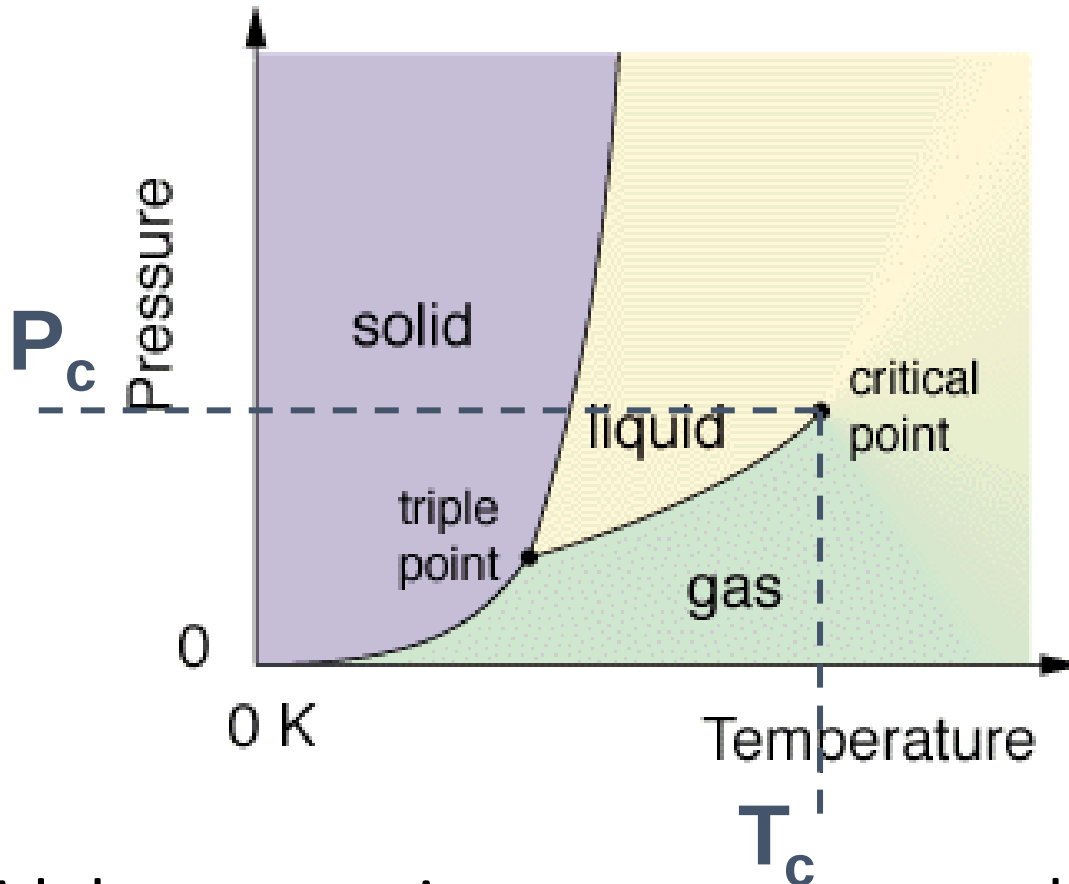


Figure 10.35 (a) A sealed container of liquid carbon dioxide slightly below its critical point is heated, resulting in (b) the formation of the supercritical fluid phase. Cooling the supercritical fluid lowers its temperature and pressure below the critical point, resulting in the reestablishment of separate liquid and gaseous phases (c and d). Colored floats illustrate differences in density between the liquid, gaseous, and supercritical fluid states. (credit: modification of work by "mrmrobin"/YouTube)

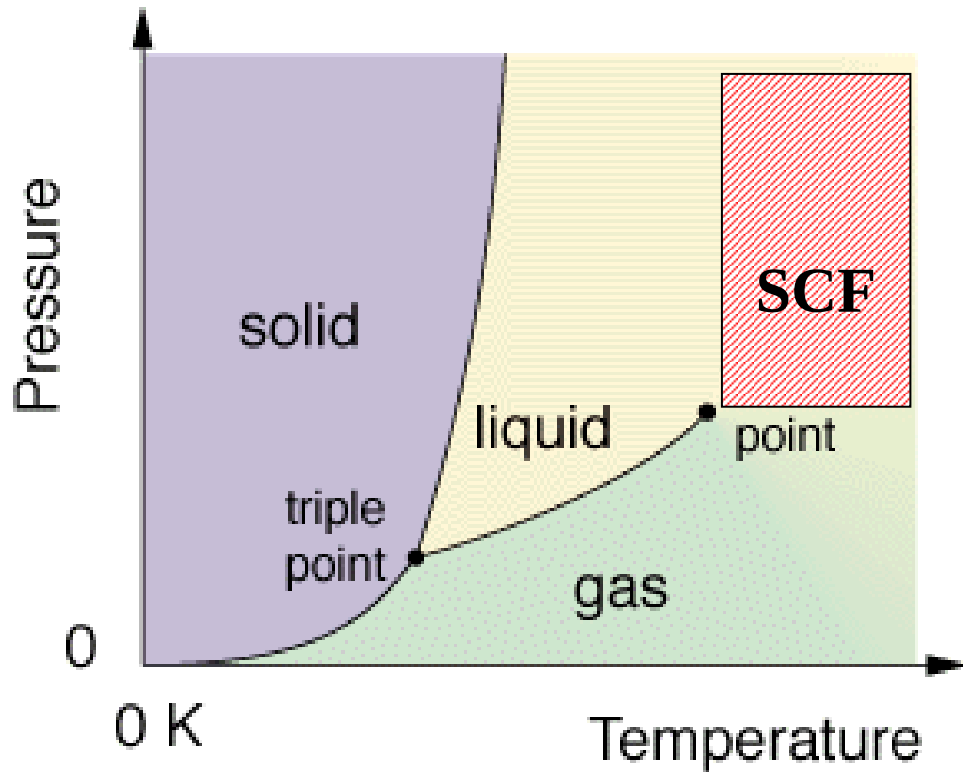
Critical Point



Substance	Critical Temperature (°C)	Critical Pressure (kPa)
hydrogen	-240.0	1300
nitrogen	-147.2	3400
oxygen	-118.9	5000
carbon dioxide	31.1	7400
ammonia	132.4	11,300
sulfur dioxide	157.2	7800
water	374.0	22,000

- A liquid does not exist at temperatures above T_c , no matter how much pressure is applied.

Supercritical Fluids



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***Ab initio* molecular-dynamics study of supercritical carbon dioxide**

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*Chemistry and Physics of Materials Unit, Jawaharlal Nehru Centre for Advanced Scientific Research,
Jakkur, Bangalore 560 064, India*

- Supercritical fluids exist above P_c and T_c
- Will expand to fill container, but dense like liquid
- Low viscosity (like a gas).
- Good solvent power (like a liquid).
- Used for extraction commercially.

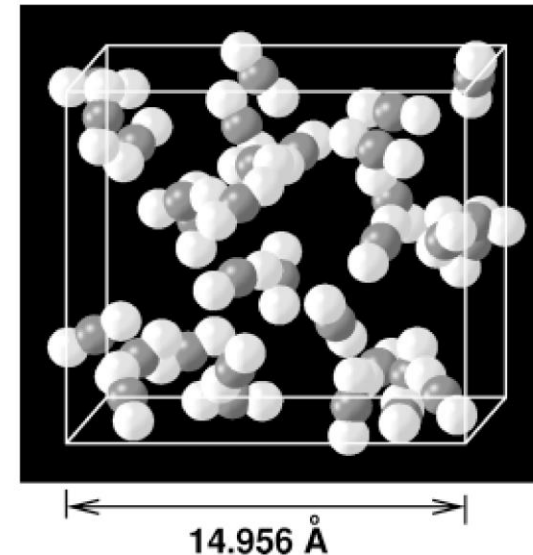
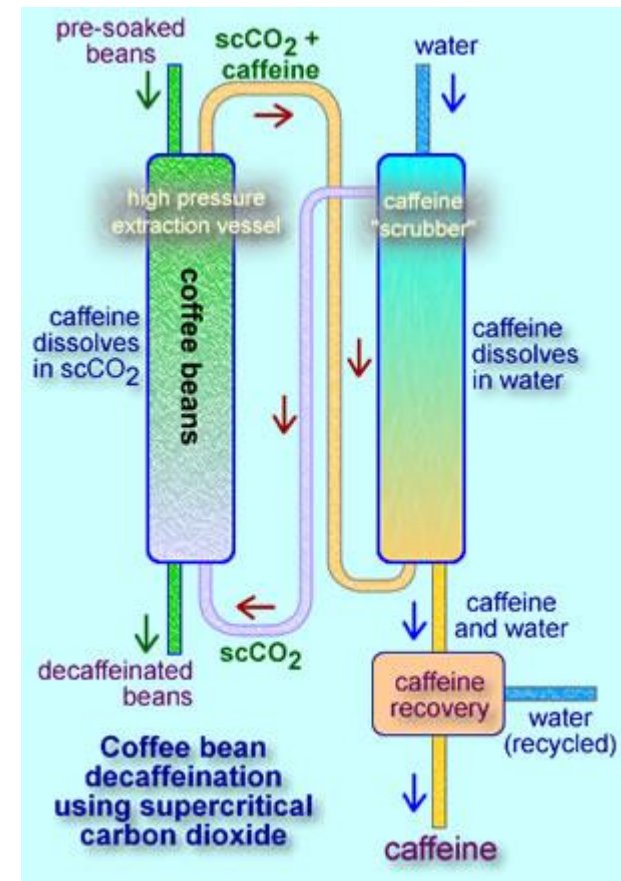


FIG. 1. Snapshot of a configuration of 32 molecules of CO_2 in the supercritical state obtained from the CPMD simulations. Oxygen atoms are shown in white, and carbon atoms are in black.



natural
product
extraction



coffee
decaffeination

Much better than the old version which used
dichloromethane